SRM Institute of Science and Technology

Delhi – Meerut Road, Sikri Kalan, Ghaziabad, Uttar Pradesh – 201204

B.Tech.-First Year, I Sem
Academic Year: 2023-24 (ODD)

Test : Internal Examination I
Course Code & Title : 21GNH101J & Philosophy of Engineering
Registration No. : RA2511030030006.

Date & Session : 13/10/2023 (AN)
Duration: 2 Hours
Max. Marks: 60

Part - A

Answer all questions **(10 x 1 = 10 Marks)**

Q. No	Question	Marks	BL	CO	PO
1	Which is the latest entry in STEAM Education ? (a) Mathematics (b) Science (c) Technology (d) Art	1	1	1	1,3
2	The great wall of China , pyramids of Egypt were part of ___era (a) Ancient era (b) Modern era (c) Middle era (d) Renaissance Era	1	1	1	1,2
3	The developments in electrical engineering were part of ___ era (a) Ancient era (b) Modern era (c) Middle era (d) Renaissance Era	1	1	1	1,3
4	Biology is an example of (a) Natural Science (b) Social Science (c) Formal Science (d) none of above	1	1	1	1,2
5	Can "teamwork" be called as a desired attribute of Engineer? (a) True (b) False	1	1	2	1,2
6	Systematic knowledge of the physical word through observation is (a) Engineering (b) Science (c) Technology (d) Art	1	1	2	1,2
7	..... is the application of science, maths and arts for the benefit of life (a) Engineering (b) Science (c) Technology (d) Art	1	1	2	1,3
8	Branch of philosophy studies existence, being, becoming and reality is (a) Engineering (b) ontology (c) Art (d) Science	1	1	2	1,2
9	Which of the following cycle is better ? (a) Open loop cycle (b) Closed loop better (c) Both (d) None	1	1	1	1,2
10	If the competition is high for a product then the product price would be (a) High (b) Low (c) Remains unaffected (d) none of above	1	1	2	1,3

Part B

Answer any five questions **6M x 5Q = 30 Marks**

11	Define Engineering.	6	1	1	1,2
12	What is the basic difference between science & technology?	6	1	2	1,3
13	What was the need to switch over to STEAM mode of education?	6	2	1	1,3
14	Discuss the major engineering disciplines in the modern era.	6	1	2	1,2
15	Which stage in the product life cycle brings maximum profit ? Why?	6	1	1	1,2
16	Give an example of mind map and explain.	6	2	2	1,2
17	What is the difference between open loop cycle and closed loop cycle.	6	2	1	1,2

This document is available on studocu
Downloaded by Madhav Gupta (mgups2002@gmail.com)

Part C

Answer all questions **10M x 2Q = 20 Marks**

18.	(A) Explain briefly about the engineering habits of mind.	10	1	1	1,2
	(OR)				
	(B) Explain briefly about Science, Engineering and Technology and their inter-relationship	10	1	1	1,2
19.	(A) Explain the product life cycle by taking any product from any brand as example	10	2	2	1,2
	(OR)				
	(B) With a neat sketch as example, explain Ontology.	10	2	2	1,3

Downloaded by Madhav Gupta (mgups2002@gmail.com)

Registration Number | R | A | 2 | 2 | 1 | 1 | 0 | 7 | 0 | 0 | 3 | 0 | 0 | 0 | 6 |

SRM Institute of Science and Technology
Department of Electrical and Electronics Engineering
Delhi – Meerut Road, Sikri Kalan, Ghaziabad, Uttar Pradesh – 201204

SRM
DELHI-NCR CAMPUS, GHAZIABAD (U.P)

Academic Year: 2023-24 (ODD)

Test : **Internal Examination II**
Course Code & Title : **21GNH101J, Philosophy of Engineering**
Year & Sem : **I Year & I Semester**

Date & Session : 1-12-2023; FN
Duration: 2.5 Hours
Max. Marks: 75

Set- S_3

Part - A

Answer all questions **(10 x 1 = 10 Marks)**

Q. No	Question	Marks	BL	CO	PO
1	What is the first step in the scientific method? A) Analysis B) Hypothesis C) Observation D) Conclusion	1	1	4	4
2	What is a hypothesis in the scientific method? A) A proven fact B) A tentative explanation for an observed phenomenon C) The final result of an experiment D) A statistical analysis of data	1	1	4	4
3	What is the first step in the engineering design process? A) Testing B) Identifying a problem or need C) Building a prototype D) Analyzing data	1	1	4	4
4	What does ADDIE stand for in the instructional design process? A) Analyze, Design, Develop, Implement, Evaluate B) Assess, Design, Deliver, Implement, Execute C) Analyze, Develop, Deploy, Integrate, Enhance D) Assess, Design, Draft, Implement, Evaluate	1	2	4	4
5	Which of the following is a key advantage of using the ADDIE model in instructional design? A) Speed and quick implementation B) Lack of flexibility C) Iterative improvement based on evaluation D) Limited consideration of learner needs	1	2	4	4
6	What is the primary focus of axiology? A) The nature of reality B) The study of knowledge C) The study of values D) The analysis of language	1	2	5	5
7	Which of the following is a sub-branch of axiology that deals specifically with moral values? A) Epistemology B) Aesthetics C) Ethics D) Metaphysics	1	2	5	5
8	Aesthetic values are concerned with the study of: A) Knowledge and truth B) Beauty and artistic experiences C) Right and wrong conduct D) Existence and reality	1	1	5	5
9	In axiology, what is the term for the study of the nature of beauty and taste? A) Ethics	1	1	5	5

This document is available on studocu
Downloaded by Madhav Gupta (mgups2002@gmail.com)

	B) Aesthetics C) Metaethics D) Epistemology				
10	Which of the following is a renewable energy source contributing to environmental sustainability? A) Coal B) Natural Gas C) Solar Power D) Nuclear Power	1	1	5	5

Part B

Answer any seven questions — **5M x 7Q = 35 Marks**

11	With the help of flow chart, explain about scientific method .	5	1	4	4
12	Explain about different operational factors of system design.	5	1	4	4
13	Explain about the contents of knowledge based on engineering.	5	3	4	4
14	What do you mean by complexity of knowledge based on engineering. Explain it with the help of a suitable example.	5	2	4	4
15	With the help of a suitable example, explain about engineering design process.	5	2	4	4
16	Explain about Engineers Code of Ethics.	5	3	5	5
17	What are the different ways to achieve Economic Sustainability?	5	1	5	5
18	Write a short note on - "Sustainable Development Goals".	5	2	5	5
19	Write a short note on - "Axiology".	5	1	5	5

Part C

Answer all questions — **10M x 3Q = 30 Marks**

20.	(A) What is ADDIE model? Explain about five different phases of ADDIE model.	10	1	4	4
	(OR)				
	(B) How Engineers contribute to the society? Explain any five areas where engineers have made a remarkable contribution to the society.	10	1	5	5
21.	(A) Describe in detail about different steps of Engineering Design Process.	10	2	4	4
	(OR)				
	(B) Discuss any five different ways to achieve environmental sustainability.	10	2	5	5
22.	(A) Explain in detail about any five engineering associations.	10	2	5	5
	(OR)				
	(B) Discuss in detail about the CDIO approach of product development with the help of a suitable example.	10	2	4	4

Downloaded by Madhav Gupta (mgups2002@gmail.com)

Reg. No	R	A	2	3	1	1	0	3	0	0	3	0	0	0	6

B.Tech/M.Tech(Integrated) DEGREE EXAMINATION, DECEMBER 2023

First Semester

21GNH101J - PHILOSOPHY OF ENGINEERING

(For the candidates admitted during the academic year 2022-2023 onwards)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks) — Answer **all** Questions

		Marks	BL	CO
1.	_____ is the discipline and profession of applying technical and scientific knowledge and utilizing natural laws and physical resources in order to design and implement materials (A) Arts (B) Mathematics (C) Engineering (D) Science	1	1	1
2.	Engineers apply the sciences of physics and mathematics to find suitable solutions to problems using_____ (A) Philosophy (B) Methodology (C) Marketability (D) Flexibility	1	1	1
3.	Engineering is considered a branch of applied _____ (A) Mathematics and science (B) Physics and mathematics (C) Imaginative and creative (D) Art and Technology	1	1	2
4.	The_____ mechanism is the earliest known model of a mechanical computer in history (A) Antikythera (B) Pneumatic (C) Phaistos (D) Cosmos	1	1	3
5.	Researchers assert that the introduction stage where design takes place determines between _____ and _____ percent of the life cycle costs. (A) 40,60 (B) 50,50 (C) 70,60 (D) 70,90	1	1	2
6.	_____describe concepts depending both on a particular domain and task (A) Application ontology (B) Reference ontology (C) Domain ontology (D) Foundational ontology	1	1	1
7.	_____ is natural extension of PLM, and creates a truly full life cycle that takes your obsolete or used products back into raw materials (A) Half loop cycle (B) Semi loop cycle (C) Open loop cycle (D) Closed loop cycle	1	1	1
8.	Ontology is sometimes referred as _____ (A) Reference (B) science of being (C) philosophy (D) Metaphysics	1	1	3
9.	_____ is creation based on the scientific knowledge put together, and technology is the set of engineered creations put together (A) Engineering (B) Science (C) Mathematics (D) Arts	1	1	2

This document is available on studocu

Downloaded by Madhav Gupta (mgups2002@gmail.com)

10. _____ is the preferred modus operandi of this dimension, where the discovery of first principles is seen as the activity leading to higher recognition
(A) Research (B) Scientific
(C) Experimental (D) Testing

11. Engineering is creating new tools, devices, and processes based on _____ knowledge
(A) Practical (B) Scientific
(C) Experimental (D) Testing

12. _____ as activity is relatèd to the conceptualization (pre-execution) stages of making new products.
(A) Design (B) Implementation
(C) Creation (D) Thinking

13. Hypothesis Testing _____ method
(A) Engineering (B) Scientific
(C) ADDIE (D) CDIO

14. The _____ of the data the system operates on is of the highest consideration when designing a reliable and fault-tolerant architecture.
(A) Security (B) Integrity
(C) Consistency (D) Reliability

15. An engineer identifies a specific need _____ need(s) _____ because _____? And then, he or she creates a solution that meets the need.
(A) Who,what,Why (B) Who,Why,What
(C) What,Why,Who (D) What,Who,Why

16. The _____model is the generic process traditionally used by instructional designers and training developers
(A) ADDIE (B) CDIO
(C) Engineering (D) Scientific

17. In _____ phase the project is reviewed and revised according to any feedback given.
(A) Development (B) Requirement Analysis
(C) Deployment (D) Testing

18. Good designers _____ possible solutions before opting to start a design, building a list of as many solutions as possible
(A) Research (B) Explore
(C) Identify (D) Brainstorm

19. Design _____ is distinct from analytic methodologies, which is crucial to develop scientific initiatives.
(A) Developers (B) Testers
(C) Thinkers (D) Epistemology

20. Design as _____ is more affiliated with management of a wide range of fields from business to military and from hospitals to academy.
(A) Engineering (B) Epistemology
(C) Planning (D) Activity

		Marks	BL	CO
10.		1	1	2
11.		1	1	1
12.		1	1	2
13.		1	1	2
14.		1	1	3
15.		1	1	1
16.		1	1	3
17.		1	1	1
18.		1	1	3
19.		1	1	3
20.		1	1	2

PART - B (4 × 10 = 40 Marks)

Answer **any 4** Questions

		Marks	BL	CO
21.	What are the different stages of engineering history? Explain the historical development of engineering with required diagrams.	10	2	2
22.	Differentiate Reference ontology and application ontology with an example.	10	3	2
23.	Compare the four dimensions of Engineering.	10	4	2

Downloaded by Madhav Gupta (mgups2002@gmail.com)

		Marks	BL	CO
24.	With neat sketch explain the engineering design process.	10	3	2
25.	what are the essential engineer role to achieve the sustainable development.	10	4	2
26.	Sketch the STEAM pyramid diagram and explain the most significant aspect of STEAM?	10	4	1

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Questions

27.	Classify the desired attributes of an Engineer with a minimum of ten key points.	15	4	1
28.	Explain the four stages of Product Life Cycle with neat sketch.	15	5	1

* * * * *

This document is available on studocu

Downloaded by Madhav Gupta (mgups2002@gmail.com)